

11.2.

$$D = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

$$D = D E D$$

$3 \times 4 \quad 3 \times 4 \quad 4 \times 3 \quad 3 \times 4$

$$\begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

$$\begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{3}a & 3b & 3c & 0 \\ \sqrt{3}d & 3e & 3f & 0 \\ \sqrt{3}g & 3h & 3i & 0 \\ \sqrt{3}j & 3k & 3l & 0 \end{pmatrix} = \begin{pmatrix} 3a & 3\sqrt{3}b & 3\sqrt{3}c & 0 \\ 3\sqrt{3}d & 9e & 9f & 0 \\ 3\sqrt{3}g & 9h & 9i & 0 \end{pmatrix}$$

$$\sqrt{3} = 3a$$

$$3\sqrt{3}b = 0$$

$$3\sqrt{3}c = 0$$

$$a = \frac{\sqrt{3}}{3}$$

$$b = 0$$

$$c = 0$$

$$3\sqrt{3}d = 0$$

$$9e = 3$$

$$9f = 0$$

$$\rightarrow E = \begin{pmatrix} \sqrt{3}/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \\ j & k & l \end{pmatrix}$$

$$d = 0$$

$$e = 1/3$$

$$f = 0$$

$$3\sqrt{3}g = 0$$

$$9h = 0$$

$$9i = 3$$

$$g = 0$$

$$h = 0$$

$$i = 1/3$$

ED je symetrická:

$$\begin{pmatrix} \sqrt{3}/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \\ j & k & l \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sqrt{3}j & 3k & 3l & 0 \end{pmatrix}$$

$$\sqrt{3}j = 0$$

$$3k = 0$$

$$3l = 0$$

$$j = 0$$

$$k = 0$$

$$l = 0$$

$$E = \begin{pmatrix} \sqrt{3}/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \\ 0 & 0 & 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$A = UDV^*$$

$$V, U \text{ orthogonálne} \rightarrow V^* = V^{-1}, U^* = U^{-1}$$

$$A = ABA$$

$$UDV^* = UDV^* B UDV^*$$

$$UDV^* V = UDV^* B UDV^* V$$

$$UD = UDV^* B UD$$

$$U^* UD = U^* UDV^* B UD$$

$$D = DV^* B UD$$

$$\text{ale } D = DED$$

$$E = V^* B U$$

$$EU^* = V^* B U U^*$$

$$EU^* = V^* B$$

$$VEU^* = VV^* B$$

$$B = VEU^*$$

$$B = \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & 2\sqrt{6} & 2\sqrt{3} \\ 0 & -3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \\ 0 & 3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \end{pmatrix} \frac{1}{3} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & 2\sqrt{3} & 2\sqrt{3} \\ -3\sqrt{2} & 3\sqrt{2} & 0 \\ \sqrt{6} & \sqrt{6} & 2\sqrt{6} \end{pmatrix} =$$

$$= \frac{1}{108} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & 2\sqrt{6} & 2\sqrt{3} \\ 0 & -3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \\ 0 & 3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \end{pmatrix} \begin{pmatrix} 6 & 6 & 6 \\ -3\sqrt{2} & 3\sqrt{2} & 0 \\ \sqrt{6} & \sqrt{6} & 2\sqrt{6} \\ 0 & 0 & 0 \end{pmatrix} = \frac{1}{108} \begin{pmatrix} 36 & 36 & 36 \\ 12 & 12 & 24 \\ 12 & -24 & -12 \\ -24 & 12 & -12 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 3 & 3 & 3 \\ 1 & 1 & 2 \\ 1 & -2 & -1 \\ -2 & 1 & -1 \end{pmatrix}$$